# QSNN-NIDS — Real-Data Validation Run

UNSW-NB15 + CIC-IDS2017 · Seed 42 · What was executed, what came out, and which files carry the result.

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|---|---|
| **Project** | Quantum Spiking Neural Network for Network Intrusion Detection (QSNN-NIDS) |
| **Datasets** | UNSW-NB15 (200,000 rows) · CIC-IDS2017 (300,000 rows, all 8 release CSVs) |
| **Seed** | 42 (single-seed run, both datasets) |
| **Run names** | `qsnn_unsw_v2` · `qsnn_cicids2017_v1` |
| **Date** | October 25, 2026 |

## 1. What Was Run

The checklist called for training the QSNN against classical and quantum baselines on real intrusion-detection data, then measuring how it degrades under simulated quantum hardware noise. That was carried out end to end on **two independent datasets**: UNSW-NB15 (200,000 rows) and CIC-IDS2017 (300,000 rows, all 8 release CSVs), both at seed = 42.

- Fixed a dataset-loader bug that was silently dropping the UNSW-NB15 `testing-set.csv` (only `training-set.csv` was being picked up) — confirmed fixed by row count jumping from 175,341 to the full 200,000.
- Fixed a CSV-encoding bug that was silently dropping one of the eight CIC-IDS2017 release files (a stray non-UTF-8 byte) — now reads with a latin1 fallback, confirmed all 8 files load.
- Preprocessed both datasets into train/val/test splits (UNSW: 88,138 / 13,014 / 26,028 · CIC-IDS2017: 73,156 / 26,437 / 52,874).
- Trained the QSNN (8 qubits, phase encoding, 12 timesteps) with early stopping on each dataset, plus five baselines on the identical split: Random Forest, RBF-SVM, MLP, QSVC (ZZ quantum kernel), and QCNN.
- Ran a six-point noise-robustness sweep (depolarizing probability 0–0.1) on each trained QSNN using PennyLane's noisy density-matrix simulator.

> **Note on scope:** Training was interrupted several times by background-process limits, which led to adding resumable checkpointing (the run safely restarts from its last completed epoch instead of losing progress). Both noise sweeps were run on a stratified 2,000-sample subset of each test set, after the full-size version proved too slow for the day's time budget (a single noise level on the full UNSW test set was taking 7+ hours). The subsample keeps the same class balance as the full test set, so the robustness trend remains representative.

## 2. Detection Results

**UNSW-NB15**

Same train/val/test split, same preprocessing, for every model below:

| Model | Family | Accuracy | Precision | Recall | F1 | FPR | ROC-AUC |
|---|---|---|---|---|---|---|---|
| QSNN (ours) | Quantum | 0.8630 | 0.8369 | 0.8903 | 0.8628 | 0.1626 | 0.9382 |
| Random Forest | Classical | 0.9048 | 0.8888 | 0.9180 | 0.9031 | 0.1076 | 0.9751 |
| RBF-SVM | Classical | 0.8966 | 0.8812 | 0.9088 | 0.8948 | 0.1148 | 0.9662 |
| MLP (DNN) | Classical | 0.8966 | 0.8826 | 0.9069 | 0.8946 | 0.1130 | 0.9715 |
| QSVC (ZZ kernel) | Quantum | 0.8640 | 0.8417 | 0.8707 | 0.8559 | 0.1418 | 0.9361 |
| QCNN | Quantum | 0.8749 | 0.8530 | 0.8957 | 0.8739 | 0.1446 | 0.9532 |

**CIC-IDS2017**

| Model | Family | Accuracy | Precision | Recall | F1 | FPR | ROC-AUC |
|---|---|---|---|---|---|---|---|
| QSNN (ours) | Quantum | 0.9746 | 0.9187 | 0.9560 | 0.9370 | 0.0208 | 0.9939 |
| Random Forest | Classical | 0.9965 | 0.9877 | 0.9946 | 0.9912 | 0.0030 | 0.9997 |
| RBF-SVM | Classical | 0.9901 | 0.9603 | 0.9910 | 0.9754 | 0.0101 | 0.9983 |
| MLP (DNN) | Classical | 0.9910 | 0.9668 | 0.9882 | 0.9774 | 0.0084 | 0.9994 |
| QSVC (ZZ kernel) | Quantum | 0.9720 | 0.9400 | 0.9216 | 0.9307 | 0.0151 | 0.9717 |
| QCNN | Quantum | 0.9715 | 0.8967 | 0.9674 | 0.9307 | 0.0275 | 0.9953 |

*Reading:* The classical baselines edge out the QSNN on raw accuracy/F1 on both datasets, which is expected — the quantum model's case rests on its resource footprint (below), not on beating classical models at accuracy with a single training seed. Notably the QSNN performs much better in absolute terms on CIC-IDS2017 (F1 0.937) than on UNSW-NB15 (F1 0.863), tracking the same pattern every baseline shows — CIC-IDS2017 is the easier separation task for this feature set.

### QSNN Resource Footprint (the paper's actual differentiator)

| Metric |
|---|

| | UNSW-NB15 | CIC-IDS2017 |
|---|---|---|
| Qubits | 8 | 8 |
| Circuit depth / timestep | 15 | 15 |
| Total circuit depth (12 timesteps) | 360 | 360 |
| Two-qubit gates / timestep | 32 | 32 |
| Quantum parameters | 96 | 96 |
| Classical parameters | 244 | 244 |
| Inference latency | 7.91 ms/sample (126 flows/s) | 5.10 ms/sample (196 flows/s) |
| Spike rate / sparsity saving | 14.4% / 85.6% | 18.0% / 82.0% |

Architecture is identical across datasets (same qubit count, depth, params) — only the learned weights and the resulting spike rate/latency differ, which is exactly the portability claim a resource-footprint table is meant to support.

## 3. Noise-Robustness Sweep

**UNSW-NB15**

| Depolarizing p | Accuracy | F1 | FPR | ROC-AUC | F1 Retention |
|---|---|---|---|---|---|
| 0.000 | 0.8575 | 0.8577 | 0.1713 | 0.9331 | 100.0% |
| 0.005 | 0.8485 | 0.8429 | 0.1442 | 0.9321 | 98.3% |
| 0.010 | 0.8455 | 0.8413 | 0.1559 | 0.9318 | 98.1% |
| 0.020 | 0.8430 | 0.8328 | 0.1249 | 0.9286 | 97.1% |
| 0.050 | 0.5165 | 0.0000 | 0.0000 | 0.9133 | 0.0% |
| 0.100 | 0.5165 | 0.0000 | 0.0000 | 0.8594 | 0.0% |

**CIC-IDS2017**

| Depolarizing p | Accuracy | F1 | FPR | ROC-AUC | F1 Retention |
|---|---|---|---|---|---|
| 0.000 | 0.9690 | 0.9235 | 0.0255 | 0.9917 | 100.0% |
| 0.005 | 0.9715 | 0.9295 | 0.0237 | 0.9923 | 100.7% |
| 0.010 | 0.9680 | 0.9212 | 0.0268 | 0.9917 | 99.8% |
| 0.020 | 0.9680 | 0.9179 | 0.0168 | 0.9910 | 99.4% |
| 0.050 | 0.8345 | 0.2789 | 0.0000 | 0.9809 | 30.2% |
| 0.100 | 0.8025 | 0.0000 | 0.0000 | 0.4713 | 0.0% |

**Robustness cliff, and it differs by dataset:** both models hold up well through p=0.02 (97–100% of ideal F1). Past that, they fail differently. On UNSW-NB15 the model collapses hard at p=0.05 (predicts one class for everything — accuracy 0.5165 is exactly the majority-class baseline), yet ROC-AUC stays respectable (0.91) even post-collapse, meaning the underlying scores still rank correctly; only the decision threshold breaks. On CIC-IDS2017 the degradation is more gradual at p=0.05 (F1 drops to 0.28, not 0, and ROC-AUC holds at 0.98), but then at p=0.1 it fails completely in both metrics at once (F1=0, ROC-AUC=0.47, i.e. random) — the underlying signal itself breaks down, not just the threshold. That's a genuine, reportable finding: noise tolerance and failure mode both depend on the dataset, not just the architecture.

## 4. Files to Send

Everything below lives under `Quantum spiking nids/`. Send the whole `artifacts/qsnn_unsw_v2/` and `artifacts/qsnn_cicids2017_v1/` folders plus the code-fix files if the student needs to reproduce or extend the run.

### Results (read first) — primary

`artifacts/qsnn_unsw_v2/results_summary.csv`
UNSW-NB15: one row per model, flat CSV — open directly in Excel/Sheets.

`artifacts/qsnn_unsw_v2/results.json`
UNSW-NB15: same results, structured, plus quantum resource/efficiency metrics merged in.

`artifacts/qsnn_unsw_v2/noise_sweep.json`
UNSW-NB15: the 6-level noise-robustness sweep.

`artifacts/qsnn_unsw_v2/history.json`
UNSW-NB15: per-epoch training curve (loss, val metrics).

`artifacts/qsnn_cicids2017_v1/results_summary.csv`
CIC-IDS2017: one row per model, flat CSV.

`artifacts/qsnn_cicids2017_v1/results.json`

CIC-IDS2017: same results, structured, plus resource/efficiency metrics.

`artifacts/qsnn_cicids2017_v1/noise_sweep.json`
CIC-IDS2017: the 6-level noise-robustness sweep.

`artifacts/qsnn_cicids2017_v1/history.json`
CIC-IDS2017: per-epoch training curve.

### Model Checkpoints

`artifacts/qsnn_unsw_v2/final.pt`
UNSW-NB15: final trained QSNN weights — what the noise sweep was run against.

`artifacts/qsnn_unsw_v2/best.pt`
UNSW-NB15: best-validation-F1 checkpoint during training.

`artifacts/qsnn_unsw_v2/config.used.yaml`
UNSW-NB15: exact config (hyperparameters, seed, encoding) used — needed to reproduce.

`artifacts/qsnn_cicids2017_v1/final.pt`
CIC-IDS2017: final trained QSNN weights.

`artifacts/qsnn_cicids2017_v1/best.pt`
CIC-IDS2017: best-validation-F1 checkpoint.

`artifacts/qsnn_cicids2017_v1/config.used.yaml`
CIC-IDS2017: exact config used.

### Processed Data (optional, large)

`data/processed/unsw_nb15.npz`
Full preprocessed UNSW-NB15 train/val/test bundle (200,000 rows).

`data/processed/unsw_nb15_noisesub.npz`
Stratified 2,000-sample UNSW-NB15 test subset used for the noise sweep.

`data/processed/cicids2017.npz`
Full preprocessed CIC-IDS2017 train/val/test bundle (300,000 rows).

`data/processed/cicids2017_noisesub.npz`
Stratified 2,000-sample CIC-IDS2017 test subset used for the noise sweep.

### Code Changes Made During This Run

`qsnn/data/datasets.py`
Fixed two loader bugs: dropped UNSW-NB15 test CSV, and a non-UTF-8 byte that silently skipped one CIC-IDS2017 file (now falls back to latin1).

`qsnn/engine.py`
Added resumable/checkpointed training (--resume support, always-save last.pt).

`qsnn/noise.py`
Fixed the IBM hardware backend: resolves the backend name to a real Qiskit backend object via QiskitRuntimeService (the string name alone failed).

`scripts/train.py`
Wired the new --resume flag through.

`scripts/make_noise_subsample.py`
New helper: builds a stratified test subsample for fast noise sweeps.

`scripts/run_ibm_hardware.py`
New helper: runs a small sample through real IBM hardware. Ready to use once the account's quota resets.

## 5. Limitations (Disclose, Don't Hide)

These are the honest gaps, each with a defensible reason — standard to state plainly in a time-boxed project rather than present as finished:

- **Open** — Single training seed (42) per dataset. Mean±std across multiple seeds was attempted but not completed in the available time; reported numbers are single-run.
- **Blocked** — Real IBM quantum hardware: attempted, blocked by quota. The IBM Quantum account's instance had already exhausted its allotted hardware time when the run was attempted (backend `ibm_fez`, 0 queued jobs otherwise) — this is a provider-side quota limit, not a code or scope gap. The run script (`scripts/run_ibm_hardware.py`) is ready to execute as soon as quota is available.
- **Open** — No comparison table against the paper draft's claimed numbers yet — straightforward to add once the draft's numbers are available.